For this assignment, you will solve numerical based on Cache Memory.

Questions

1. Consider a single-level cache with an access time of 2.5 ns, a line size of 64 bytes, and a hit ratio of $H = 0.95$. Main memory uses a block transfer capability that has a first-word (4 bytes) access time of 50 ns and an access time of 5 ns for each word thereafter.
   - **a)** What is the access time when there is a cache miss? Assume that the cache waits until the line has been fetched from main memory and then re-executes for a hit.
   - **b)** Suppose that increasing the line size to 128 bytes increases the H to 0.97. Does this reduce the average memory access time?

2. Consider a cache of 64 Kbytes. Data is transferred between main memory and cache in blocks of 4 bytes each. The main memory consists of 16 Mbytes, with each bit directly addressable by a 24-bit address. For mapping purposes consider main memory to consist of 4M blocks of 4 bytes each. Show the address format for direct mapping, associative mapping and 2-way set associative mapping.

3. Use the address formats from Q2. For the hexadecimal main memory addresses 111111, 666666, BBBBBB, show the following information, in hexadecimal format:
   - **a)** Tag, Line, and Word values for a direct-mapped cache.
   - **b)** Tag and Word values for an associative cache.
   - **c)** Tag, Set, and Word values for a two-way set-associative cache.

4. A set-associative cache consists of 64 lines, or slots, divided into four-line sets. Main memory contains 4K blocks of 128 words each. Show the format of main memory addresses.

5. A two-way set-associative cache has lines of 16 bytes and a total size of 8 Kbytes. The 64-Mbyte main memory is byte addressable. Show the format of main memory addresses.

6. Consider a machine with a byte addressable main memory of 2^{16} bytes and block size of 8 bytes. Assume that a direct mapped cache consisting of 32 lines is used with this machine.
   - **a)** How is a 16-bit memory address divided into tag, line number, and word number?
   - **b)** Suppose the byte with address 0001 1010 0001 1010 is stored in the cache. What are the addresses of the other bytes stored along with it?
   - **c)** Into what line would bytes with each of the following addresses be stored?
     - i. 0001 0001 0001 1011
     - ii. 1100 0011 0011 0100

7. Consider a machine with a byte addressable main memory of 2^{16} bytes and block size of 8 bytes. Assume that a two-way set associative cache consisting of 32 lines is used with this machine.
   - **a)** How is a 16-bit memory address divided into tag, set number, and word number?
   - **b)** Into what set would bytes with each of the following addresses be stored?
     - i. 1101 0000 0001 1101
     - ii. 1010 1010 1010 1010

8. Consider a cache with a line size of 32 bytes and a main memory that requires 30 ns to transfer a 4-byte word. For any line that is written at least once before being swapped out of the cache, what is the average number of times that the line must be written before being swapped out for a write-back cache to be more efficient that a write-through cache?

9. Consider an L1 cache with an access time of 1 ns and a hit ratio of H = 0.95. Suppose that we can change the cache design (size of cache, cache organization) such that we increase H to 0.97, but increase access time to 1.5 ns. What conditions must be met for this change to result in improved performance?